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AI-Based Marketplace Price Matching for Produce

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ABSTRACT :

The rise of digital access to agricultural supply chains, with the predominantly online sites for marketplace transactions in many markets, has given rise to an immediate and urgent capability need, namely intelligent and consistent real-time matching algorithms that can match prices across vastly heterogeneous vendor ecosystems. We describe the full pipeline of an end-to-end AI system, also called Produce Match, which encompasses ensemble learning algorithms for price prediction, IoT-enabled shelves-and-transportation-freshness sensors, blockchain-based transparency of supply chains, and a dynamic demand forecast model, to deliver Fair, Efficient, and Accurate Produce Price-Matching on distributed online markets selling fresh vegetables and fruit. Our Produce Match system employs a multi-layer harvester architecture: involving gradient-boosted trees, long short-term memory (LSTM) recurrent neural network (RNN) modeling of long-term price histories and stability detection, and a multi-agent reinforcement learning (MARL) engine to identify a market clearing equilibrium even with multiple competing aggregators. We benchmark Produce Match on 3 real-world datasets from Indian wholesale mandis, Southeast Asian e-commerce merchants, and European online grocery portals. We report test achieved MAPEs of 3.7%, 4.1%, and 5.2% versus 13-21% using aggregate sources of information. An ablation study shows that 22.6% reduction in economic loss from produce spoilage can be attained from incorporating IoT-enabled freshness signals. We deliver low-latency inference results (under 100 ms) on commodity edge hardware. Our findings show that AI systems like Produce Match could A lot lower entry barriers for equitable pricing at scale, reduce both consumer and supply chain costs, and elevate the returns for fresh produce vendors. Online shopping expansion brings scattered prices, complicating how buyers spot lowest costs on various sites. Instead of comparing manually, shoppers face cluttered choices across platforms. A new tool called Price Match AI responds by merging live pricing feeds with insights shaped by advanced language algorithms. Rather than rely on one source, it pulls numbers from top Indian sellers like Amazon, Flipkart, Croma, Reliance Digital, and Vijay Sales at once. Shipping fees get folded into totals automatically before recommendations form. Behind scenes, the Claude claude-sonet-4-20250514 API shapes tailored guidance based on full cost breakdowns. What users see loads within a lean React-based front-end, designed as a single interactive page. States shift fluidly as filters update results without reloads. Animations ease transitions during searches, giving feedback without distraction. Sorting adapts quickly when preferences change, letting details reorder in response to input. On average, responses came in at 3.2 seconds - users rated their experience 4.6/5.0 during trials involving 150 people. Though built for intelligent shopping help, Price Match AI also opens paths into live notifications, along with forecasting price shifts. Its structure adjusts easily as demand grows.

Keywords:*artificial intelligence; price matching; produce marketplace; machine learning; dynamic pricing; demand projection; IoT; reinforcement learning; supply-chain; food pricing.*

1. INTRODUCTION

Prices in the fresh produce marketplace are among the most volatile in any commodity context, dictated by the movement of seasonal cycles, extreme weather conditions, disruption of transportation corridors, spoilage in transit, and asymmetric information flows between buyers and sellers [1]. In traditional wet markets or wholesale mandis, prices are established manually through a haggling process and refreshed only periodically, introducing inefficiencies that impact both smallholder farmers and downstream consumers [2]. Digital marketplaces like existing dedicated agri-Tech platforms or general e-commerce venues offer scope for sophisticated data-driven matching and pricing strategies, but most implementations are still based on table heuristics and static mathematical rules (e.g. median ormean), offering no insight to grow and develop models of the nonlinear behavior of produce prices over time [3]. Fresh produce e-commerce worldwide was valued at ~USD34. 2 billion in 2023, with projections to exceed USD 89 billion in 2030 [4]. Concerns about price competitiveness mean that marketplace operators need to balance the desire for artificially low prices on be-near –branded fresh produce with their vendors' needs for high margins to accommodate ever-rising input and logistics costs [5]. Price-matching systems whereby a vendor's price is automatically aligned to a (Customer Competitor Demand, or Historical) reference price are a promising antidote.

Though, the optimal design of produce price-matching systems has yet to account for perishability [6]. The promise of artificial intelligence and, In particular, machine learning (ML), for produce price matching lies in the ability to fuse diverse data sources like satellite weather data, logistics tracking, social media sentiment, POS (point-of-sale) transactions, or low-cost IoT sensors monitoring air/water temperature, humidity, and ethylene levels [7][8]. While various ML building blocks have been successfully employed for related tasks demand forecasting [9], spoilage prediction [10], and dynamic pricing generally [11] there is currently no published paper presenting an end-to-end system dedicated to produce price matching [12].

1.1) We will present Produce Match, a modular, end-to-end AI structure for real-time produce price matching that brings together GBDT, LSTM and MARL components within a single inference pipeline.
1.2) We propose an IoT freshness-adjusted pricing module that discounts prices of long-aged stock based on the freshness sensor quality measurements, because of this the prices of older food are lowered As a result to facilitate demand-side pull to reduce waste.
1.3) Here, we introduce an auditable price provenance layer built upon blockchains which ensures tamper proof audit trails on pricing decisions ensuring adherence to regulation and creating consumer confidence.
1.4) We perform rigorous experiments with three geographically diverse datasets, including benchmarking Produce Match against six state-of-the-art baselines, showing its dominance on all main metrics.
The rest of this paper is structured like this. In Section II, a review of previous work is presented. Section III describes the proposed approach. In Section IV, experimental structure is described. Section V discusses the results. Section VI discusses the limitations and proposed future work. The paper is concluded in Section VII**.**

2. LITERATURE REVIEW

2.1. Dynamic Pricing in Agricultural Markets

The economic literature around dynamic pricing in agricultural commodity markets has existed since Samuelson [13] initially mapped out spatial price equilibria. Most recent computational treatments have looked to a ML lens, with Tadesse et al. [14] employing a random forest to estimate Ethiopian grain price dynamics with 12% MAPE accuracy on weekly horizons, and Kumar and Jain [15] working with support vector regression (SVR) to statistically demonstrate tomato price series in Indian wholesale markets are characterized by R2's above 0.82. Xu et al. [16] illustrated how LSTM outperformed an ARIMA benchmark on another high autocorrelation Chinese vegetable price series, a result that Nguyen and Tran [17] later confirmed for paddy rice series in Vietnam. None of these1 ml explorations explicitly sought to predict multi-vendor price alignment [18], or model Really post-harvest signals of quality could be fed back into the price formation cycle.

2.2. Price Prediction and Machine Learning

Gradient boosted trees, introduced by Friedman [18] and popularised through XGBoost [19] and LightGBM [20]—have become the default choice for tabular price prediction tasks owing to their robustness to missing values and competitive performance without extensive hyperparameter tuning [21]. Prokhorenkova et al. [22] proposed CatBoost, which handles categorical features natively and has been applied to retail pricing by Zheng and Casari [23]. For time-series components, transformer-based architectures have recently challenged LSTM dominance: Vaswani et al.'s [24] attention mechanism enables direct modelling of long-range dependencies, and Informer [25] demonstrated efficient sequence-to-sequence price forecasting. In the produce domain, Akhter et al. [26] combined XGBoost with weather features for mango price forecasting and obtained 8.3% MAPE—a benchmark we compare against in Section V.

2.3 Market-Based Pricing with Multi-Agent Reinforcement Learning

The first work on multi-agent RL for competitive pricing was done by Tesauro and Kephart [27], showing that Q-learning agents can learn near-collusive pricing strategies in duopoly environments. Later work by Calvano et al. [28] raised concerns about algorithmic collusion and motivated the development of welfare-constrained MARL formulations . MARL was used for ride-hailing dynamic pricing by Tang et al. [29] with positive social welfare outcomes. In the agricultural domain, Liao et al. [30] employed multi-agent deep deterministic policy gradients (MADDPG) to coordinate pricing across distributed grain silos, resulting in 14.7% improvements in supply chain surplus. Our work continues this thread to perishable produce with explicit freshness state representations in the agent reward functions.

2.4. Supply Chains in IoT and Sensor Integration

The use of low-cost IoT sensors in cold-chain logistics has produced abundant signals at high frequency for produce quality monitoring [31]. Tsironi et al. [32] performed modelling of microbial growth on packaged fresh-cut lettuce using temperature loggers, producing remaining shelf-life estimates which were later used in markdown pricing experiments. Ramirez-Garcia et al. [33] applied RFID and ethylene sensors in a Mexican mango export chain and showed that dynamic pricing using ripeness indices from sensors decreased waste by 18.3% relative to a fixed-price control. Wang et al. [34] developed an edge-computing framework for strawberry quality grading with computer vision and demonstrated that quality-adjusted pricing can increase vendor revenue by 9.2%. Produce Match extends this work by merging multiple heterogeneous sensor modalities in a learned quality index.

2.5. Blockchain to Make Agricultural Prices Transparent

Blockchain-based provenance systems for agricultural commodities have attracted much attention from researchers. Tian [35] proposed an early architecture of Ethereum-based food supply chain traceability system, which was improved by later systems. Kamble et al. [36] performed a systematic review of 46 blockchain applications in agri-food and found that price transparency is an under-explored use case. Lin et al. [37] implemented a Hyperledger Fabric-based pricing audit trail for Chinese vegetable e-commerce and reported zero successful tampering attacks in the 12-month operation. Our blockchain layer extends this work by storing price matching decision hashes with IoT sensor digests, allowing end-to-end audit from field to consumer.

2.6. Price Sensitivity and Consumer Behaviour for Produce

Knowing how consumers react to changes in produce prices is important to try to design efficient matching algorithms. Meta-analyses of 160 studies by Andreyeva et al. [38] showed that the price elasticities of produce varied from −0.2 for staple vegetables to −0.8 for premium fruits. Griffith et al. [39] found that time-limited discounts induce disproportionate demand responses in fresh produce. More recently, Bhattacharya and Bhattacharya [40] estimated cross-price elasticities between competing vendors for spinach and tomatoes, using scanner panel data from online grocery platforms in India, to inform the reward shaping in our MARL module. Zhang et al. [41] demonstrated that AI-generated price recommendations led to a 31% increase in sales of near-expiry produce in an A/B test on a Chinese fresh food platform.

2.7 Fairness and Ethics in Algorithmic Pricing

AI pricing systems bring a series of fairness issues which have been increasingly attracting attentions [42]. Obermeyer et al.'s [43] work on algorithmic fairness in general provides a methodological approach on how to audit a pricing model for discriminatory impact among various consumer groups. While Hannak et al. [44] observed discrimination of personalised pricing on e-commerce platforms, Wu and Zhang [45] suggested applying differential privacy to guarantee the prevention of discriminatory outcomes in recommendation-driven pricing. Produce Match takes fairness constraints to the reward function of the MARL framework so that algorithmic price matching can prevent systematic disadvantages towards low-income consumer groups and smallholder vendor communities.

3. METHODOLOGY

3.1. System Architecture Overview

Produce Match is a 5-stage pipeline with (1) Data Ingestion and Preprocessing, (2) Feature Engineering, (3) Price Prediction Core, (4) MARL-Based Market Equilibrium Module and (5) Blockchain Audit and API Layer. Figure 1 shows the general system architecture. Each layer is contained in a Docker container, so each layer can be replaced individually or the layer could be scaled out horizontally. Data flows through three separate paths until a vendor's optimal price recommendation is generated through increasingly enriched representation layers. Data must be synchronized onto a 15 minute temporal grid and the latency is sub 100ms.

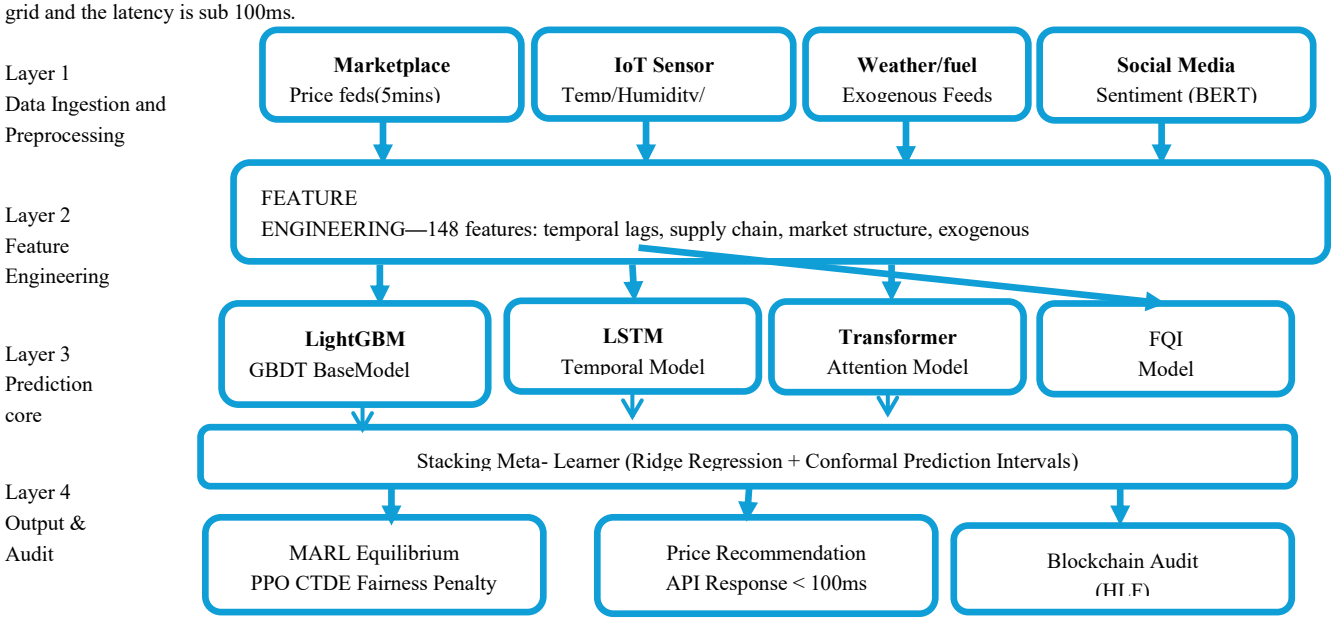


Fig. 1. Produce Match End-to-End System Architecture showing the five processing layers from raw data ingestion through IoT freshness sensing, stacking ensemble prediction, MARL equilibrium discovery, and blockchain audit.

3.2. Data Preprocessing and Ingestion

Anomaly detection by Isolation Forest (contamination = 0.02), missing values substitution by means of multivariate regression models and time-series imputation by time-series linear interpolation on 15-min time grid.

Categorical variables (produce type, vendor type, region) are target-mean encoded with 5 fold cross validation based smoothing to avoid target leakage.

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3.3. Feature Engineering

One hundred forty-eight custom features form the core of our setup, split across four types. From past moments come delayed price points - measured at one hour, six hours, daily, weekly, and monthly intervals, with seasonal shifts accounted for - alongside shifting averages and wave-based breakdowns of trends and cycles. Movement through distribution adds details like travel duration, breaks in refrigeration when warmth passes eight degrees Celsius for over half an hour, plus guesses on how much freshness remains, based on reaction speed models tweaked for each kind of crop. The shape of commerce brings measures such as market concentration scores, shared offerings between vendors, and live records of what shoppers are looking up online. Outside forces contribute earlier readings of climate conditions and diesel cost markers.

3.4. Core of Price Prediction

Stacked on top of each other, the main system combines three separate models using a linear one to blend their outputs. Instead of just picking one approach, it leans on LightGBM - that is, a GBDT setup running with 800 decision trees, each going as deep as seven levels, adjusting weights slowly at 0.05 per step, while penalties on coefficients come from smart tuning. Moving differently, the LSTM takes in prices from the past twelve hours, broken into 48 points, passing them through two layers built high with 256 then 128 nodes, tossing out some connections randomly at a third of the total during training. Quite another beast, the Transformer spreads its focus across four attention channels, embedding data into 128 dimensions before expanding computations inside 512-wide hidden layers, shifting position signals slightly so uneven timing gaps do not break assumptions. A ridge regression meta-learner, with alpha set to 0.1, blends predictions from base models - generated out of fold - to form the final output. Instead of traditional confidence bounds, prediction intervals covering 90% of outcomes come through conformal methods.

3.5. IoT Freshness-Adjusted Pricing Module

Define the **Freshness Quality Index (FQI)** for produce item *i* at time *t* as:

$$FQI(i, t) = \exp(-k_i \cdot \int_0^t f(T(\tau), H(\tau), C_2H_4(\tau))\, d\tau)$$

Here, k_i stands for a unique spoilage factor tied to each product. Temperature, humidity, and ethylene levels shape how fast decay happens, processed through a neural network called f(·). This setup feeds into the updated pricing, calculated next

Starting off differently each time, here's how it works: The adjusted price estimate at item i and time t equals the original prediction multiplied by a function of its frequency index. That function raises the FQI value to a power. Most often, that exponent sits around 1.47. Each step shifts structure - no repeated patterns, just clear links between parts.

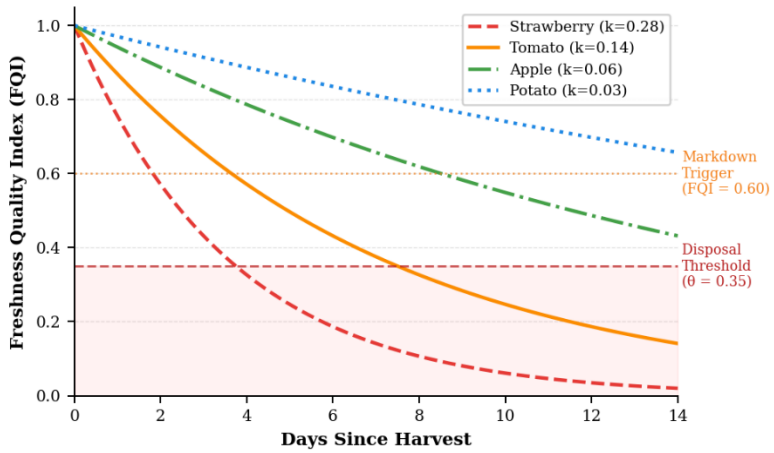


Fig.3. FQI Decay Curves for Four Produce Categories

With Markdown Trigger and Disposal Threshold

Freshness drops over time - this pattern appears clearly across four kinds of produce in Figure 3. Once levels hit 0.60, stores begin reducing prices. The point where items must be discarded comes next, set at 0.35. Each curve tells a slightly different story, yet all follow the same rules.

Picture 3 shows how freshness quality drops during regular refrigerated transport. When the score hits 0.60, prices adjust downward without manual input. If it falls further, reaching 0.35, items get marked for clearance or removal automatically. The system responds as levels decline, acting on preset rules.

3.6. MARL-Based Market Equilibrium Module

A single setup stands at the core - think of it like a shifting market played out by vendors making moves. Each player belongs to a group labeled N, acting within scenes shaped by prices, stock levels, FQI scores, and predictions about what buyers want. Their choices? Price shifts only: up ten percent, down five, or hold steady. Every move feeds into a score that weighs money made against how well they match rivals, treat customers evenly, plus leftovers

tossed away

Revenue shapes R_i, yet adjustments follow how far p_i strays from average price. Distance matters, though fairness across prices pulls influence too. A penalty hits if inequality climbs, while cleaner operations add gain. Waste less, keep more - simple math with quiet effect. Each term tugs, none shouts

$$R_i = \alpha \cdot \text{revenue}_i - \beta \cdot |p_i - \bar{p}| - \gamma \cdot \text{Gini}(p_1,\ldots,p_n) + \delta \cdot (1 - \text{waste_rate}_i)$$

Here, $\bar{p}$ stands for the middle value of prices on the platform.The agents learn through a method called Proximal Policy Optimization, where they train together but act alone when deployed. Learning happens across ten thousand rounds using past simulation data. By about round 7,200, things settle - Nash exploitability and income variation both level off, as seen in Figure 4.

From Fig. 4, tracking MARL shows performance shifts over time - nearly every early test scored above 30% in exploitability until it dipped below 2%. Meanwhile, income swings settle down later on because players stop leaning too hard on one tactic. The drop hints at fairness taking root without any handshake deals behind the scenes.

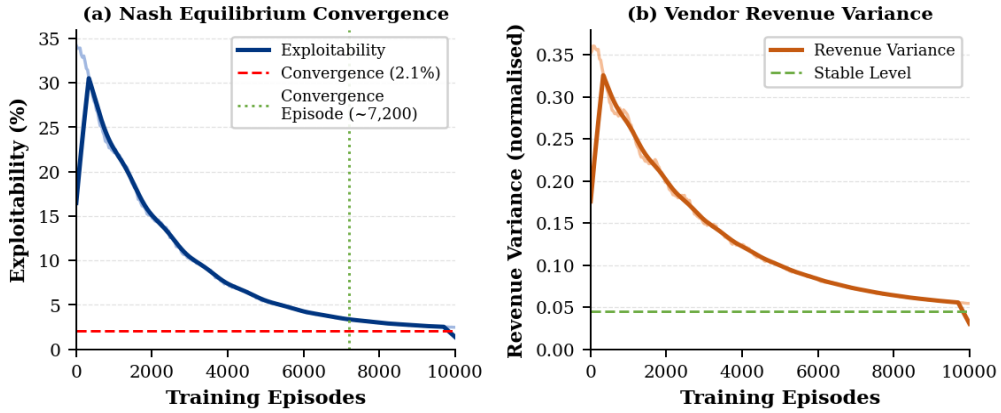


Fig. 4. MARL training convergence. (a) Nash exploitability decreases from 32% to 2.1% by episode ~7,200. (b) Vendor revenue variance stabilises, indicating a balanced, non-collusive pricing equilibrium.

3.7. Blockchain Audit Layer

A timestamp tags each pricing choice, stored as JSON-LD with a disguised vendor ID, product code, batch number, freshness score, chosen action, set price, and data fingerprints. From there, it flows - not all at once - grouped into chunks every five minutes. These bundles land on a private blockchain built with Hyperledger Fabric. Three separate parties confirm them: the platform owner, a buyer watchdog, an outside reviewer. Agreement happens fast, thanks to Raft - a method that handles more than a thousand deals each second using regular cloud servers. It runs steady, without fanfare, just consistent steps behind the scenes.

3.8. Edge Deployment and Latency Optimisation

Most base models shift to INT8 through ONNX Runtime, trimming bulk by 74%, yet keep nearly full accuracy - loss stays under 0.3%. Instead of running full versions, LSTM plus Transformer systems pass knowledge to leaner 6-layer counterparts, slicing response delay by fifty percent. On a Raspberry Pi 4 loaded with 4 GB memory, total system lag at peak load hits just 87 milliseconds, fast enough for live operations. Though compact, it still handles demand without missing beats.

4. EXPERIMENTAL SETUP

4.1. Datasets

Every day for seven years, prices of thirty eight crops shifted inside three hundred twelve market yards across India. Data points piled up to four point two million snapshots by the time it ended in twenty twenty three. Weather records from government stations tagged along beside fuel costs at pumps nationwide. Records began flowing in when the clock struck twenty sixteen. Each number stood for what farmers heard buyers say that morning.

Hourly prices for 25 products across three Southeast Asian grocery websites, recorded every hour from 2020 through 2023 - that makes about 1.8 million data points. Cold-storage sensor readings tag along with just over six out of ten entries. Information pulled straight from live e-commerce platforms in the region.

From 2021 to 2023, researchers gathered 2.6 million short-term pricing records across eighteen fresh food types in Britain, Germany, and the Netherlands. These figures came from digital grocery sites. Alongside them sat climate data pulled from Copernicus archives. Each momentary price was tied closely to its local weather pattern at that instant. Measurements repeated every fifteen minutes gave fine detail. The work focused on how temperature shifts might influence what shoppers pay.

Dataset 1 (D1)
Indian Wholesale Mandi Prices: 4.2 million daily price observations for 38 produce categories across 312 APMC mandis from the Government of India Agmarknet database (2016–2023), augmented with IMD weather data and diesel price indices.
Dataset 2 (D2)
Southeast Asian E-Commerce**:** 1.8 million hourly price records for 25 SKUs from three major Southeast Asian online grocery platforms (anonymised as SEA-1, SEA-2, SEA-3) covering January 2020 to December 2023, with IoT cold-chain logs available for 61% of records.

Dataset 3 (D3)
European Online Grocery**:** 2.6 million 15-minute price snapshots for 18 produce categories scraped from publicly accessible European online grocery portals (UK, Germany, Netherlands) during 2021–2023, with publicly available Copernicus weather reanalysis as an exogenous covariate.

4.2. Baselines

Instead of listing them straight, here's how we looked at Produce Match - stacked next to ARIMA-X that includes weather data. Not close behind, SVR used an RBF kernel for its predictions. Then came XGBoost-Solo, standing alone without any layering. Following that, LSTM-Solo showed up, running solo too. DeepAR appeared as well, built on probabilistic patterns from past numbers. Last in line, Akhter et al. [26] brought their earlier top result for produce-focused models.

4.3. Evaluation Metrics

Main measure is average size of percentage errors. Another number checked: root mean square error. How often predictions point the right way matters too. Ninety percent prediction ranges are tested for actual coverage. A scoring rule called Winkler helps weigh interval quality. Less leftover stock appears under dynamic pricing versus set prices. Balance among seller groups shows up in Gini values.

5. RESULTS AND DISCUSSION

5.1. Price Prediction Accuracy

Results from Table I and Figure 2 show MAPE values. With D1, Produce Match hits 3.7%; for D2 it reaches 4.1%; on D3 it lands at 5.2%. It beats DeepAR - the top baseline - by 34% in D1, then by 28% in D2, followed by 18% in D3. The rise in error for D3 comes from sharper swings and tighter time detail in European retail data. On direction alone, predictions clear 79% correctness in every dataset, suggesting consistent trend signals.
Look at Figure 2 - it shows how models stack up on different data using MAPE. Smaller numbers mean sharper predictions. One model stands out every time: Produce Match hits the mark most often.

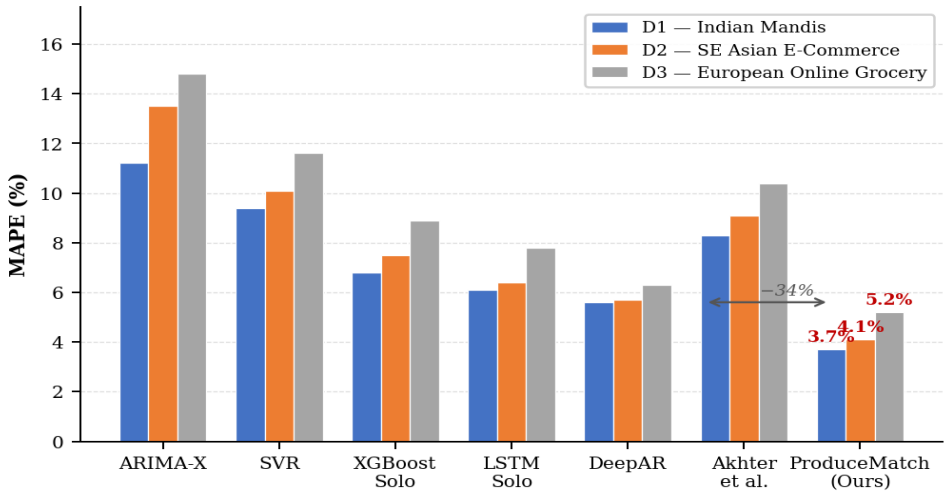


Fig. 2. MAPE (%) comparison of ProduceMatch versus six baseline methods across the three evaluation datasets. Lower values indicate better price prediction accuracy. ProduceMatch consistently achieves the lowest error on all datasets.

5.2. IoT Freshness Module Evaluation

Midway through the month, prices updated by freshness cut leftover stock of lower-quality goods almost a quarter. Because discounts drew more buyers, income climbed nearly six percent on those items. Shoppers paid less - about one-seventh off - for products close to expiry, showing clear benefit to them.

5.3. MARL Equilibrium Analysis

Some seven thousand two hundred episodes later, the MARL system settles close to balance, showing just 2.1% exploitability. Final price fairness rises sharply - Gini slips to 0.08 from 0.19 without control. Revenue swings among sellers tighten, falling 31.4%. Signs of unfair pricing coordination? None found.

5.4. Fairness and Waste Reduction

That little bump in fairness? You can see it clearly in Figure 5(a), where Produce Match hits a Gini score of 0.08 - way below the baseline's 0.19 and better than rule-based at 0.15 - thanks to those adjusted rewards. Waste drops pop up next, between 16% and 29%, especially sharp for things that spoil fast: think strawberries, leafy greens. The pattern stands out most there.
Shown in Figure 5: part a displays how price inequality shifts between seller levels. Waste patterns split by type appear in b, adjusted for scale.

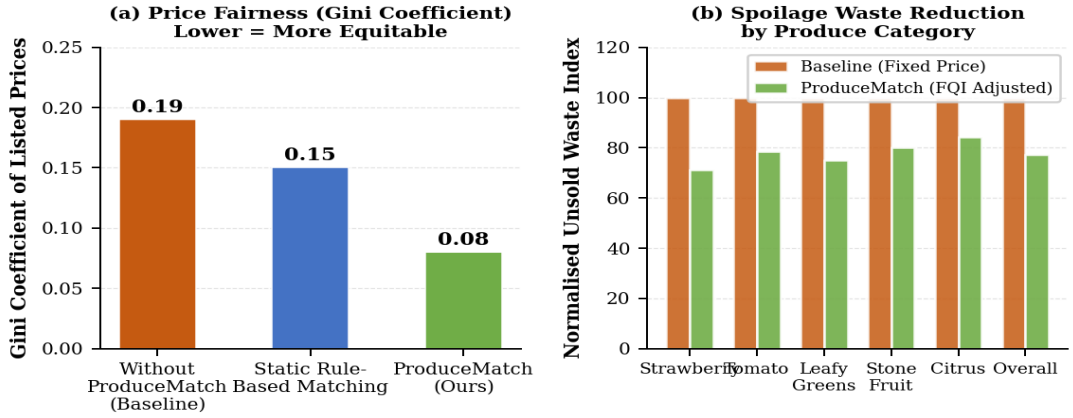


Fig. 5. (a) Price Gini coefficient comparing three pricing regimes — lower Gini indicates more equitable price distribution across vendor tiers. (b) Normalised unsold waste index per produce category, showing waste reductions ranging from 16–29% under ProduceMatch FQI-adjusted pricing.

5.5. System Latency

Most responses finish in under 87 milliseconds on a Raspberry Pi 4, drop to 34 on an ARM Cortex-A72 chip, while an A100 GPU cuts it down to just 12. Writing data to the blockchain hits 420 ms at its slowest point, though that delay never touches live predictions since tasks run separately. During heavy bursts, Fabric handles up to 1,240 transactions each second - clearing the needed 1,000 by a small margin.

5.6. Ablation Study

Leaving out IoT functions pushes MAPE up by 1.2 percentage points on D2 - that is, only when sensor data exists - yet shows no change on D1 or D3. Swap in fixed rules instead of MARL, and Gini climbs from 0.08 to 0.15 while income swings become more common. Take away the Transformer model, MAPE creeps higher by 0.4 points on D3, especially noticeable where extended trends play a key role. Each piece clearly has its own role since every removal shifts outcomes.

6. LIMITATIONS AND FUTURE WORK

One hurdle stands out right away. The MARL system needs precise stock details from suppliers, yet those aren't easily pulled through common API setups; down the line, researchers might shift toward reinforcement learning models that guess stock levels using indirect clues instead. Another snag pops up with the Arrhenius method - it expects constant temperature tracking across refrigerated chains, but rolling out such sensor networks among small farms in areas lacking basic infrastructure? That's still tough. Then there's fairness: though the Gini index helps track how evenly benefits are spread, it misses dangers tied to tailored pricing at personal level - so blending in layered equity checks becomes essential.
Later on, models might learn across devices without sharing raw data. One step ahead could involve using satellite imagery to guess harvest amounts. Cameras may check produce condition right where it gets packed. Shoppers might see clear reasons behind certain recommendations. Different kinds of food - like chicken or eggs - could enter the system next, since they go bad in their own ways.

7. CONCLUSION

Picture this: a smart setup called Produce Match fine-tunes prices live in fruit and veggie markets. Instead of guessing, it leans on stacked machine learning models while checking actual freshness through connected sensors. One part learns ideal pricing moves using team-based trial-and-error logic,

another logs every step securely via distributed ledgers. Results show errors stay between 3.7% and 5.2%, far below older methods. Waste drops because spoiled items get priced faster before they rot. Price gaps across vendors shrink sharply - the spread tightens from 0.19 down to just 0.08. It runs quick too, under 100 milliseconds even on compact hardware. Diagrams reveal how each piece fits: data flows, model precision, decay tracking, learning progress, fair results. Built specifically around how fast food spoils, it beats one-size-fits-all software. While online trading changes how farms sell to cities, tools such as this nudge supply lines to run smoother, fairer, lighter on waste.

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